

The Relaxation of Molecular Spin Qubits

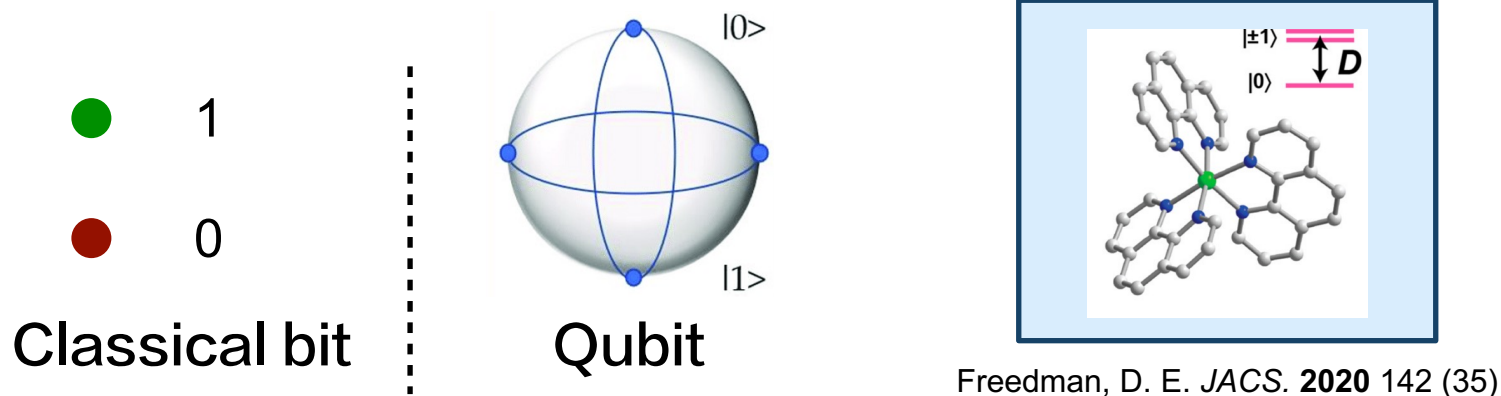
Temperature dependence and the “spin-phonon” polaron mechanism

Rui-Hao Bi
Westlake University

The 1st Degree of Freedom Series Workshop 20260718

Molecular Spin Qubits

- Molecular spin systems in quantum information

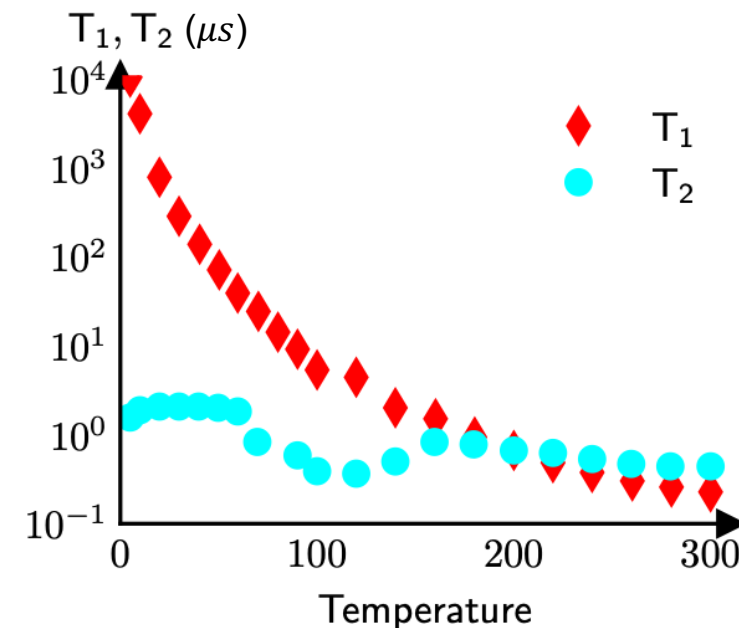
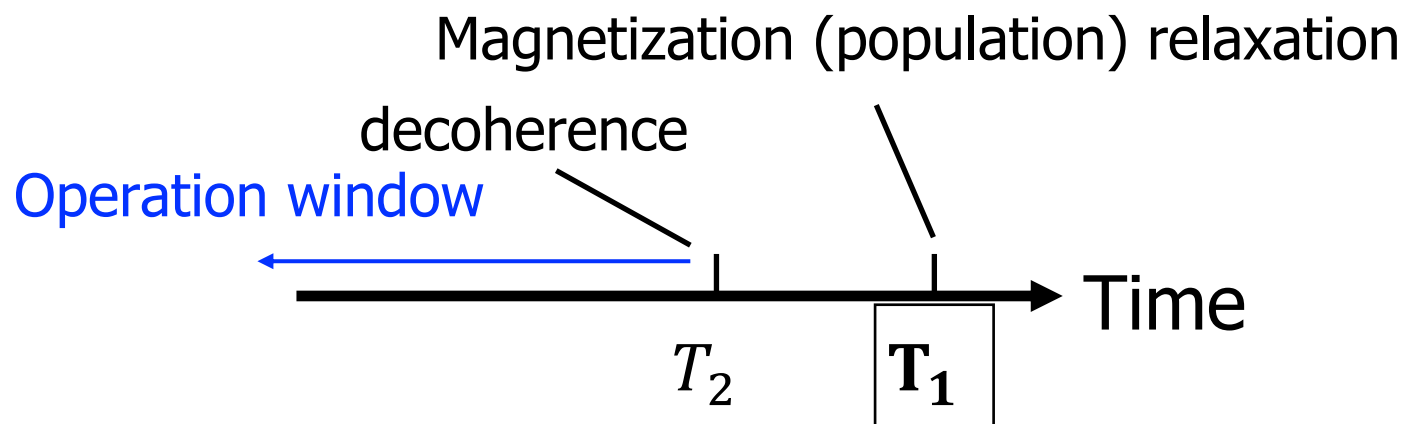


Scalable (synthesis)

Chemical modulation

...

- The spin-lattice relaxation time T_1



Challenges emerge for classical theories

- Classical perturbation theories used to fit experimental data

PHYSICAL REVIEW VOLUME 129, NUMBER 6 15 MARCH 1963

Direct Spin-Lattice Relaxation Processes*

J. W. CULVAHOUSE, WESLEY P. UNRUH,[†] AND DAVID K. BRICE[‡]

Spin-lattice relaxation in rare-earth salts

BY R. ORBACH^{†‡}
Clarendon Laboratory, University of Oxford

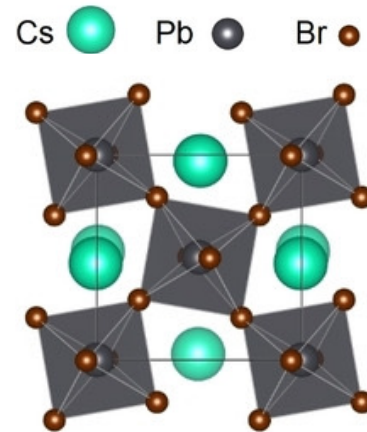
PHYSICAL REVIEW VOLUME 130, NUMBER 1 1 APRIL 1963

Raman Spin-Lattice Relaxation of Shallow Donors in Silicon

THEODORE G. CASTNER, JR.

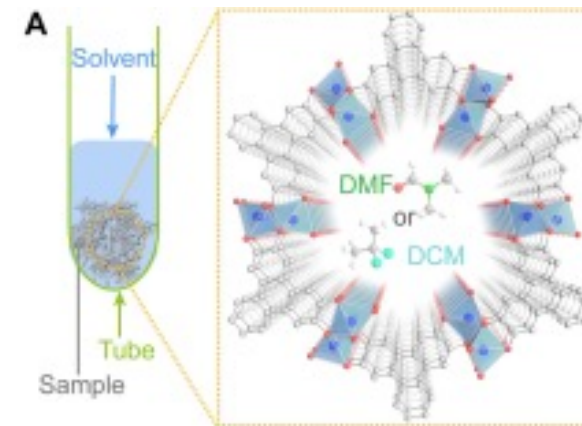
$$\frac{1}{T_1} = A_{\text{dir}} T + A_{\text{Ram}} \left(\frac{T}{\theta_D} \right)^9 J_8 \left(\frac{\theta_D}{T} \right) + A_{\text{Orb}} \left[\frac{\Delta_{\text{Orb}}^3}{\exp(\Delta_{\text{orb}}/T) - 1} \right] + \dots$$

- Challenges



Strong spin-phonon couplings

Ping, Y. *Nat Commun* **2024** 15 (1)



Complex vibrational environment

Lei, S. *arXiv:2506.04885* **2025**

A minimal two-level system model for spin relaxation

- Zeeman Hamiltonian for a molecular spin qubit:

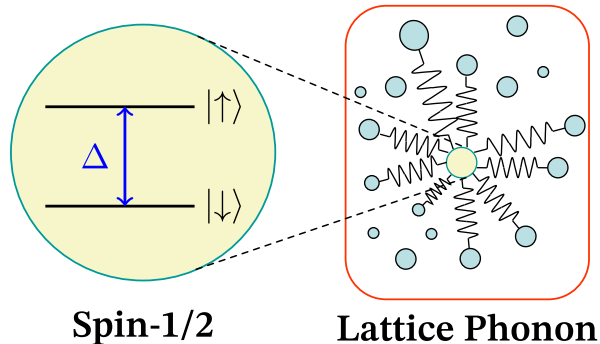
$$H_s = \mu_B (B_x \ B_y \ B_z) \cdot \mathbf{g}(\mathbf{Q}) \cdot (\sigma_x \ \sigma_y \ \sigma_z)^T \quad \text{Landé g-tensor.}$$

Lunghi, A.; and Sanvito, S. *Sci. Adv.* **2019** 5 (9)

- Spin-phonon coupling

$$H_{\text{sph}} = \mu_B \mathbf{B} \cdot \left[\sum_{\alpha \mathbf{q}} \left(\frac{\partial \mathbf{g}}{\partial Q_{\alpha \mathbf{q}}} \right) Q_{\alpha \mathbf{q}} + \sum_{\substack{\alpha, \beta \\ \mathbf{q}, \mathbf{q}'}} \left(\frac{\partial^2 \mathbf{g}}{\partial Q_{\alpha \mathbf{q}} \partial Q_{\beta \mathbf{q}'}} \right) Q_{\alpha \mathbf{q}} Q_{\beta \mathbf{q}'} + \dots \right] \cdot \boldsymbol{\sigma}$$

- A simpler spin-boson model



$$H = H_s + H_{\text{ph}} + H_{\text{sph}}$$

$$H_{\text{sph}} = (\alpha_1 \hat{Q} + \alpha_2 \hat{Q}^2 + \dots) V_{\text{sph}} \left\{ \begin{array}{l} \sigma_z, \\ \sigma_x, \sigma_y \end{array} \right.$$

$$\hat{Q} = \sum_j c_j (b_j^\dagger + b_j)$$

Onizhuk, M.; and Galli, G. *Rev. Mod. Phys.* **2025** 97 (2)

Methodology: FGR and the extended DEOM.

- Fermi's Golden Rule (FGR)

$$T_1^{-1} = k_{1 \leftarrow 0} + k_{0 \leftarrow 1},$$

$$k_{f \leftarrow i} = \frac{2\pi}{\hbar} \sum_a P_a \sum_b |W_{f,b \leftarrow i,a}|^2 \delta(\Delta).$$

Nitzan, A. *JCP* **1974** 60 (10);

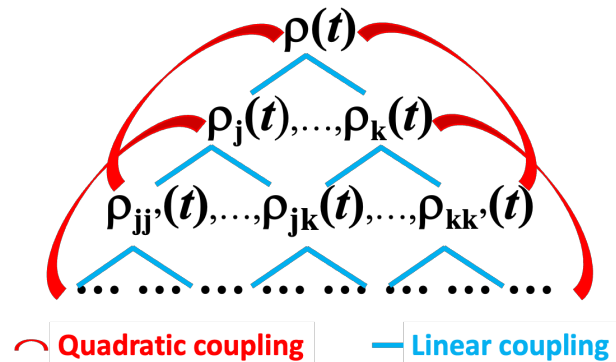
Sanvito, S. *Sci. Adv.* **2019** 5 (9);

Dou, W. *JCP* **2024** 161 (2)

$$H_{\text{sph}} = (\alpha_1 \hat{Q} + \alpha_2 \hat{Q}^2 + \dots) V_{\text{sph}}$$

One-phonon Two-phonon Multi-phonon

- The extended DEOM



$$k_{f \leftarrow i} = \int_0^\infty d\tau \underbrace{\mathcal{K}_{f \leftarrow i}(\tau)}_{\text{Rate kernel}}$$

Yan, Y. *Front. Phys.* **2016** 11 (4);

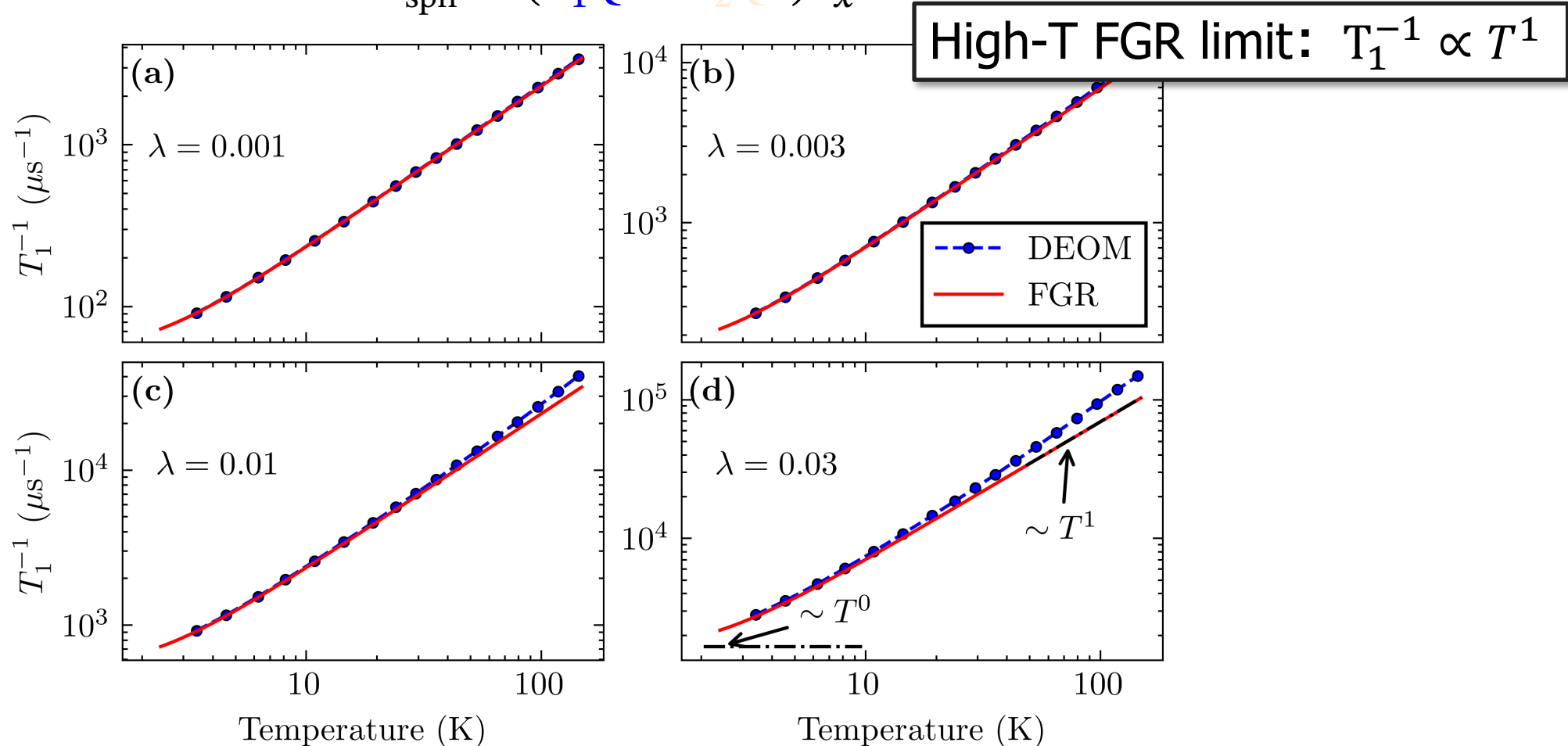
Yan, Y. *JCP* **2018** 148 (11);

Dou, W. *JCP* **2024** 161 (2)

Study 1: The failure of FGR at strong coupling

Single-phonon process : FGR underestimates T_1^{-1}

$$H_{\text{sph}} = (\alpha_1 \hat{Q} + \alpha_2 \hat{Q}^2) \sigma_x$$

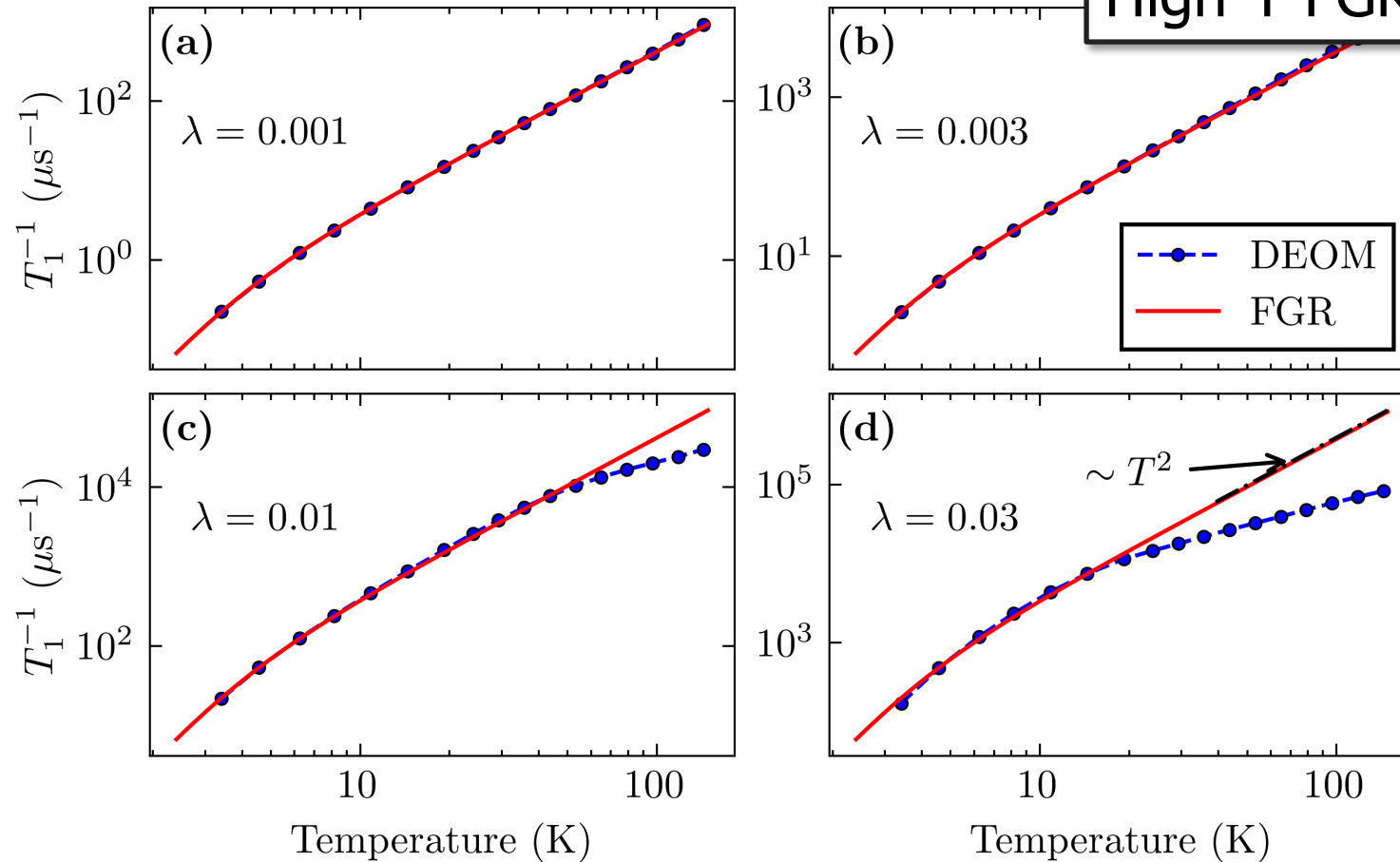


Study 1: The failure of FGR at strong coupling

Twp-phonon process : FGR over estimates T_1^{-1}

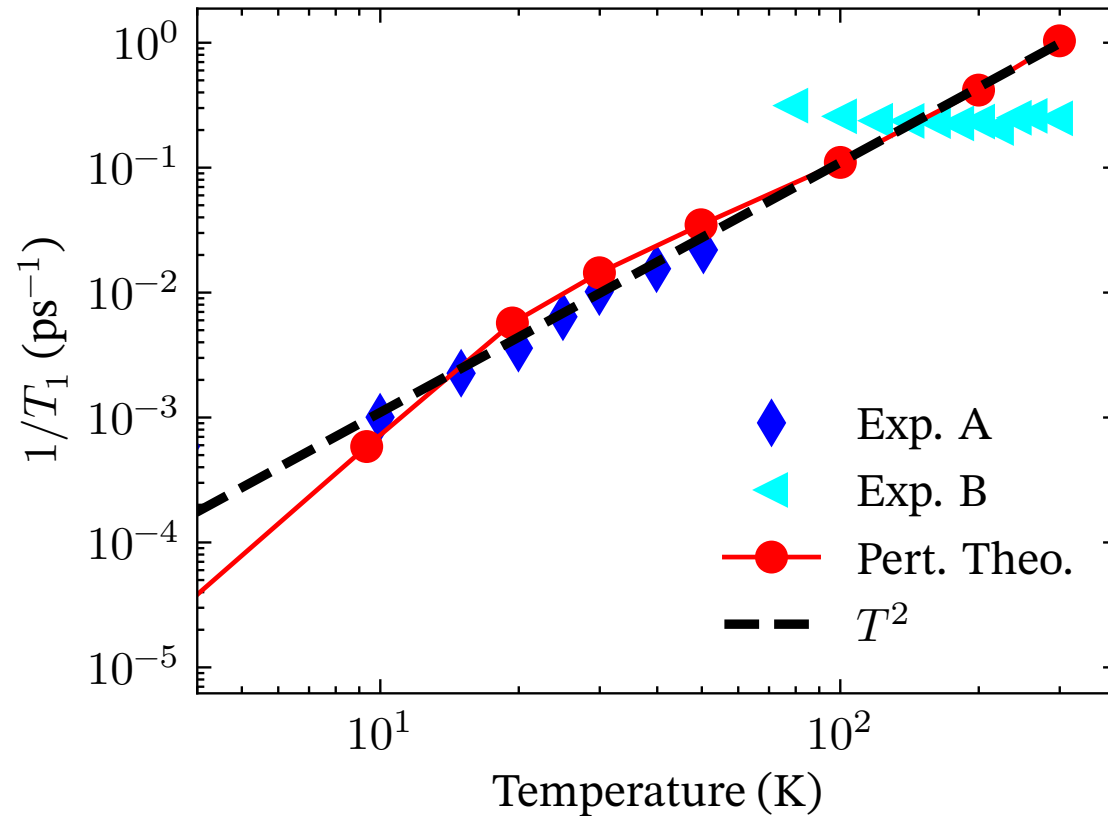
$$H_{\text{sph}} = (\alpha_1 \hat{Q} + \alpha_2 \hat{Q}^2) \sigma_x$$

High-T FGR limit: $T_1^{-1} \propto T^2$

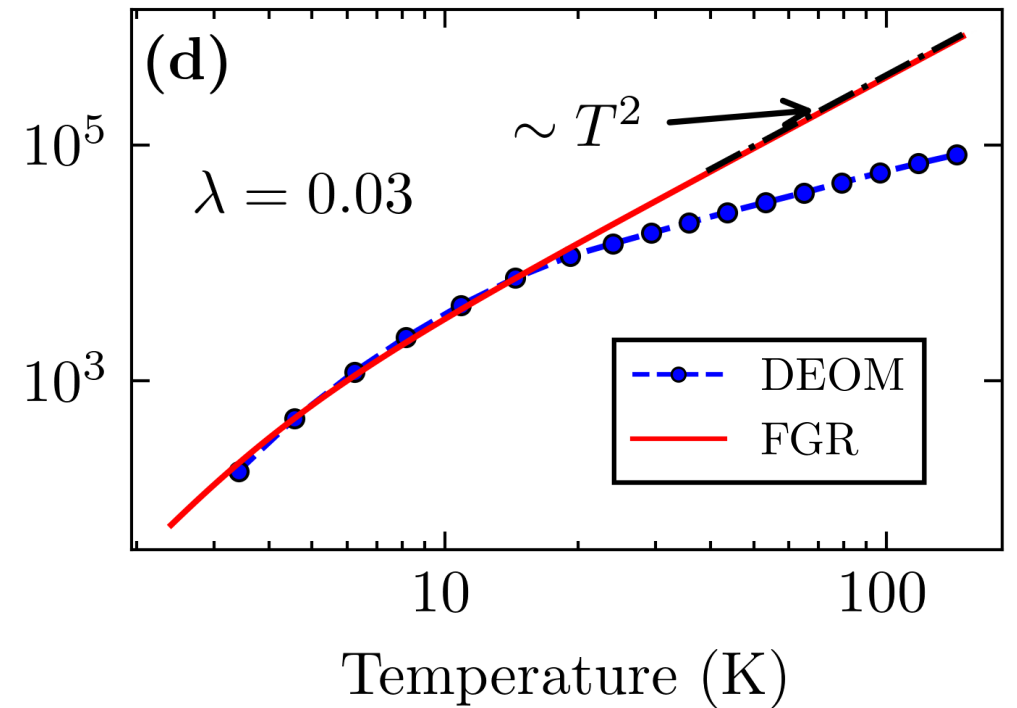


Study 1: The failure of FGR at strong coupling

Same discrepancy of $T_1^{-1} \propto T^2$ found in perovskite CsPbBr₃



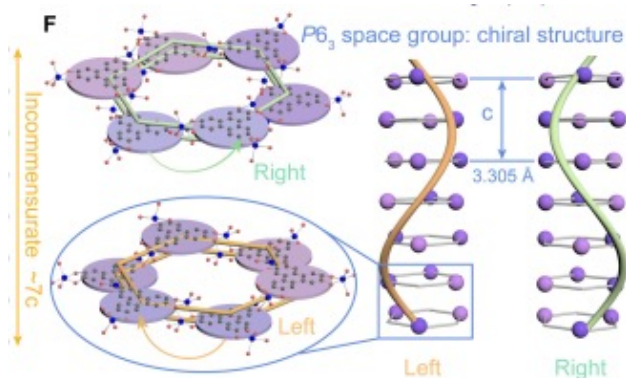
Replotted from Ping, Y. *Nat Commun* **2024** 15 (1)



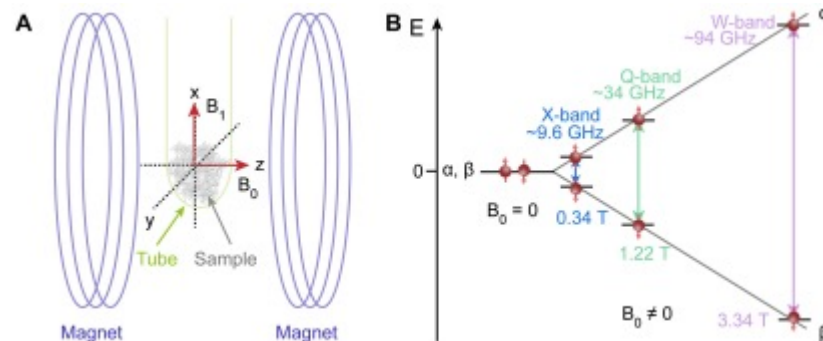
Dou, W. *JCP* **2024** 161 (2)

Study 2: anomalous relaxation in $\text{Zn}_3(\text{HOTP})$

MOF $\text{Zn}_3(\text{HOTP})$

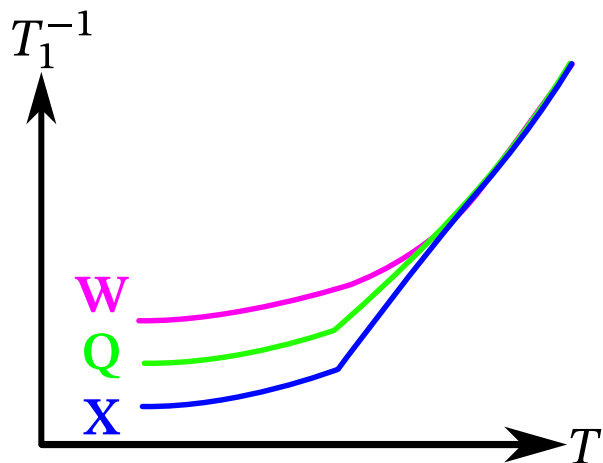


EPR setup

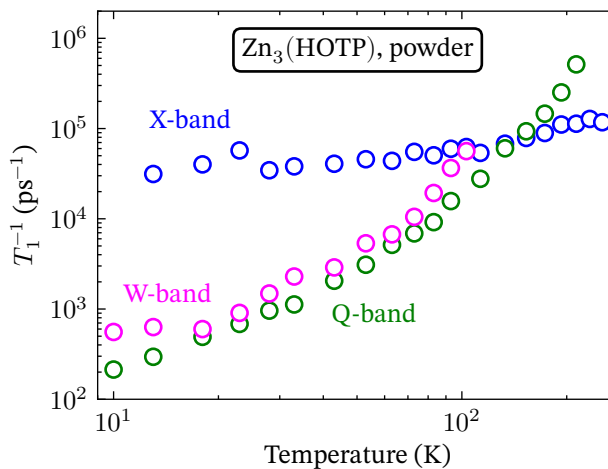


Lei Sun @ Westlake

Classical theories



Experiments



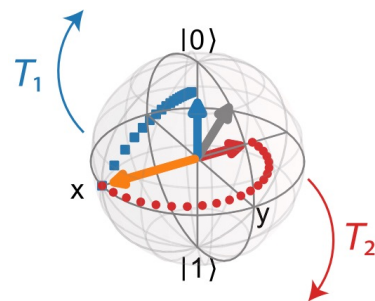
- **Efficient** relaxation in X-band
- Increase external field $\rightarrow$ back to "normal".

Study 2: anomalous relaxation in $\text{Zn}_3(\text{HOTP})$

Traditional picture

$$H_{\text{sph}} = (\alpha_1 \hat{Q} + \alpha_2 \hat{Q}^2 + \dots) V_{\text{sph}}$$

$$V_{\text{sph}} \propto \begin{cases} \sigma_x & \text{Relaxation, } T_1. \\ \sigma_z & \text{Decoherence, } T_2 \end{cases}$$



Only consider transverse coupling: $V_{\text{sph}} \propto \sigma_x$

“Spin-phonon polaron” picture

$$H_{\text{sph}} = (\alpha_1 \hat{Q}_r + \alpha_2 \hat{Q}_r^2) \sigma_x + \gamma_p \hat{Q}_{\text{sp}} \sigma_z$$

“relaxation” modes

Small polaron transform

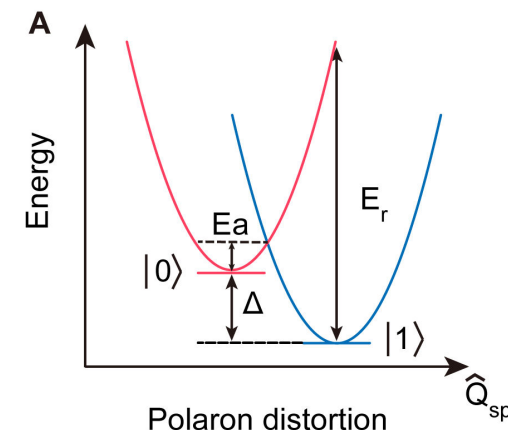
“modulation modes”

$$\tilde{H}_{\text{sph}} = (\alpha_1 \hat{Q}_r + \alpha_2 \hat{Q}_r^2) \exp\left(i \sum_k \bar{\lambda}_k \hat{p}_{k,\text{sp}}\right) \sigma_+ + \text{h.c.}$$

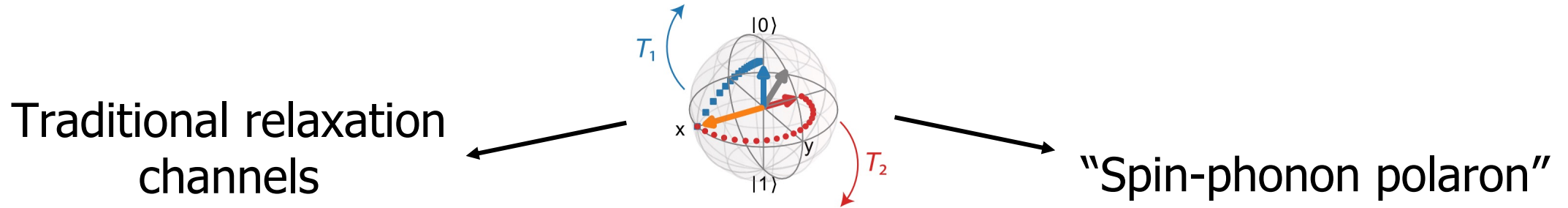
- “Spin-phonon polaron”: FGR results

$$T_1^{-1}(\text{with polaron}) \propto T_1^{-1} \exp\left[-\frac{(\Delta - E_r)^2}{4k_B T E_r}\right]$$

$$\begin{aligned} \Delta &\propto B, \\ E_r &\propto B^2 \end{aligned}$$



Study 2: anomalous relaxation in $\text{Zn}_3(\text{HOTP})$

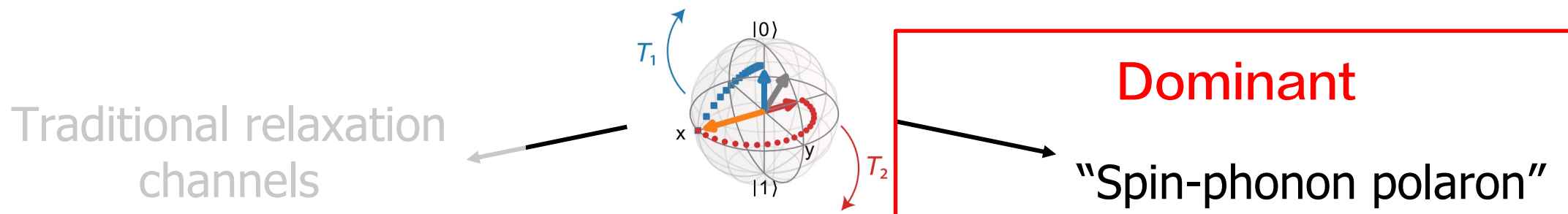


Zero-field

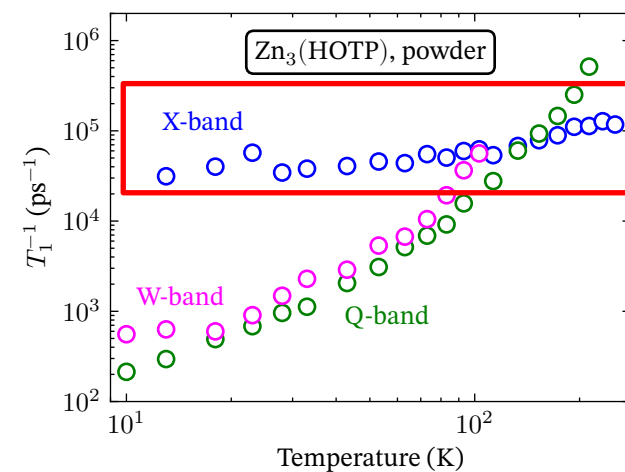
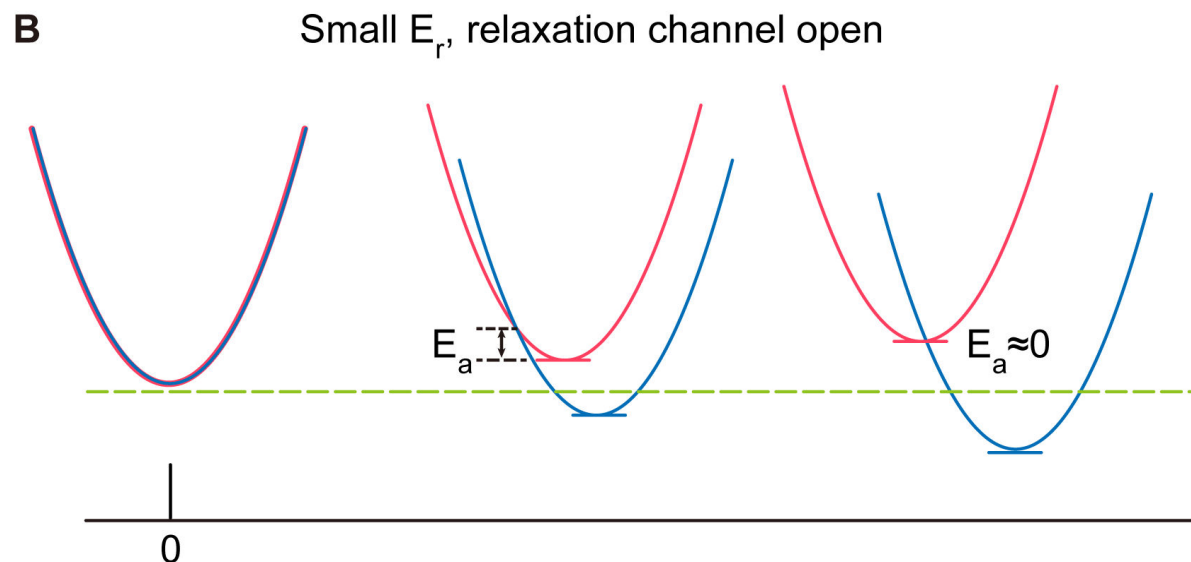
B



Study 2: anomalous relaxation in $\text{Zn}_3(\text{HOTP})$



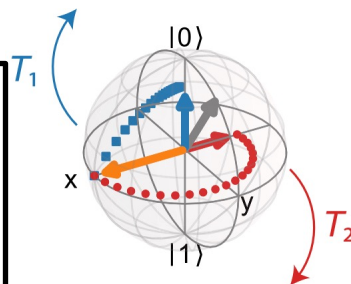
Intermediate field strength



External field B

Study 2: anomalous relaxation in $\text{Zn}_3(\text{HOTP})$

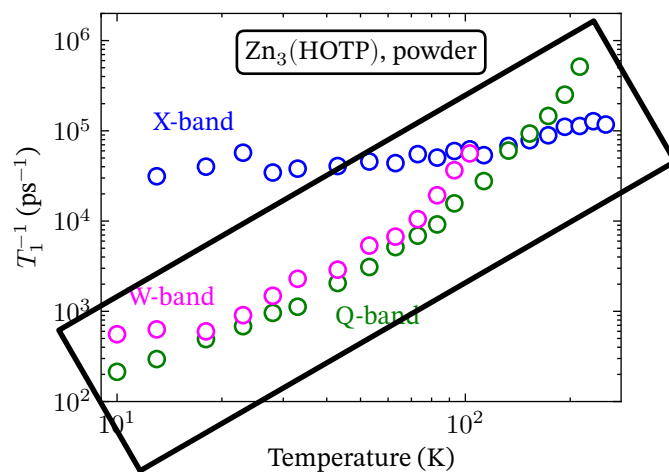
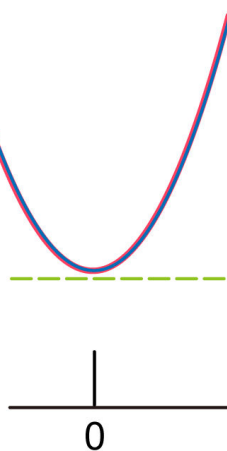
Dominant
Traditional
relaxation channels



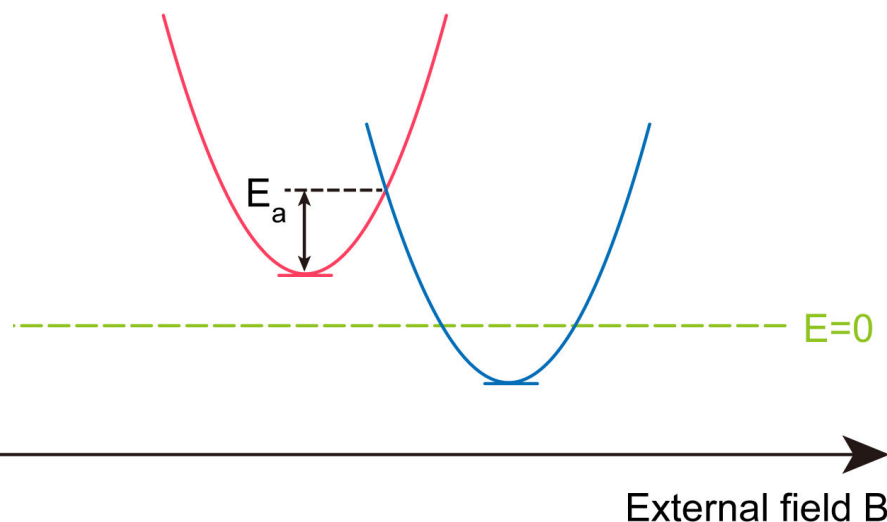
“Spin-phonon polaron”

High field strength

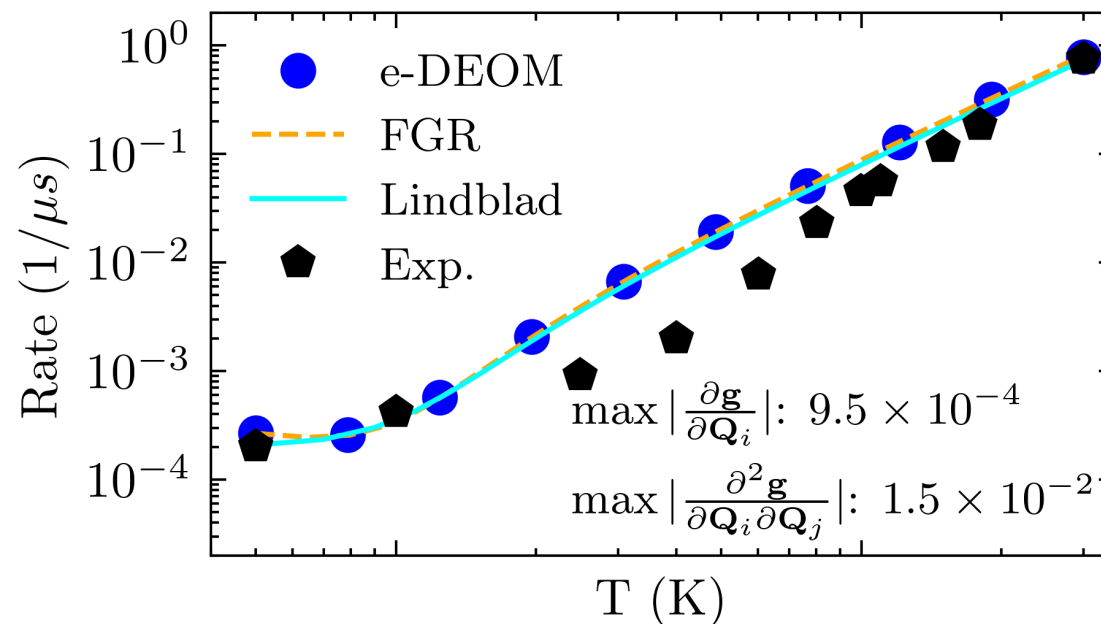
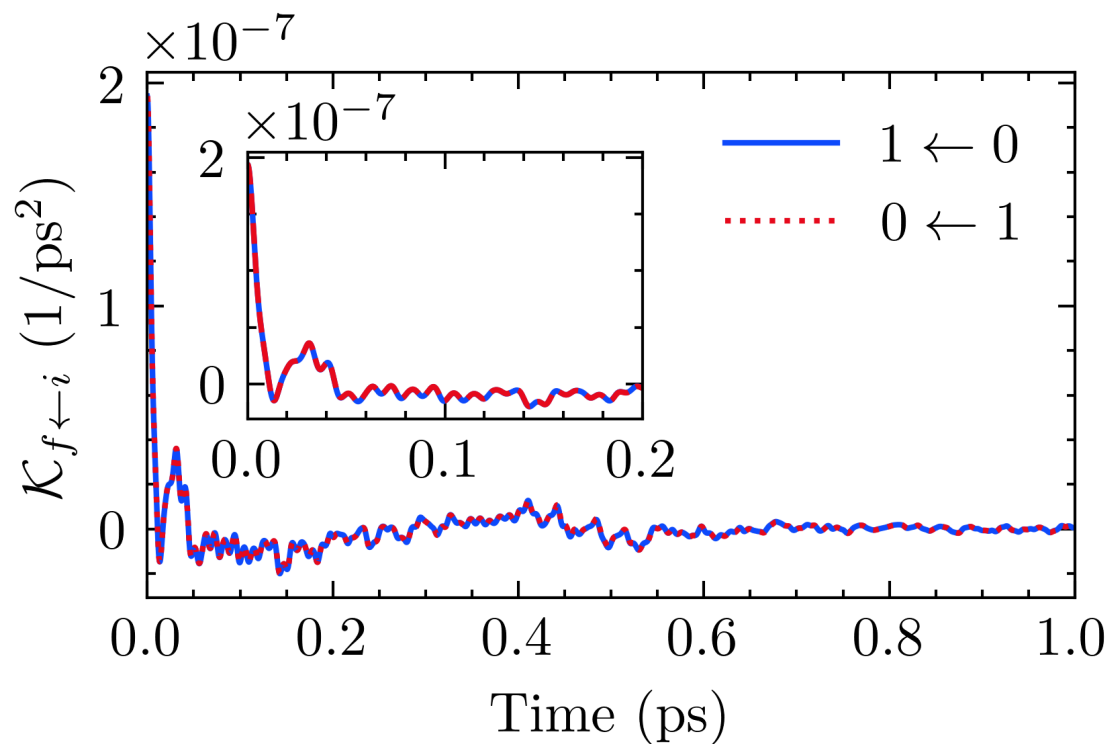
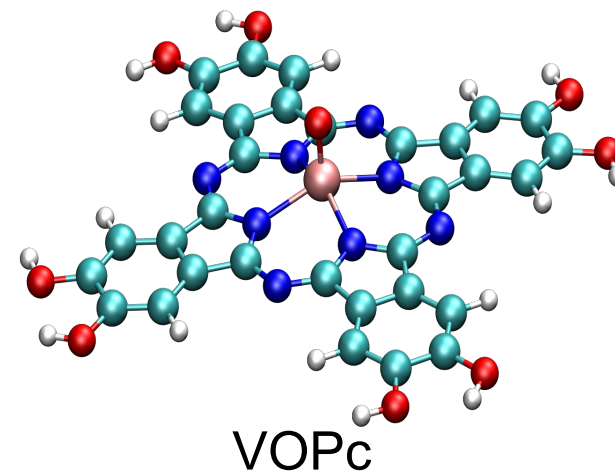
B



Much larger E_r , relaxation channel closed



Application to realistic molecular systems

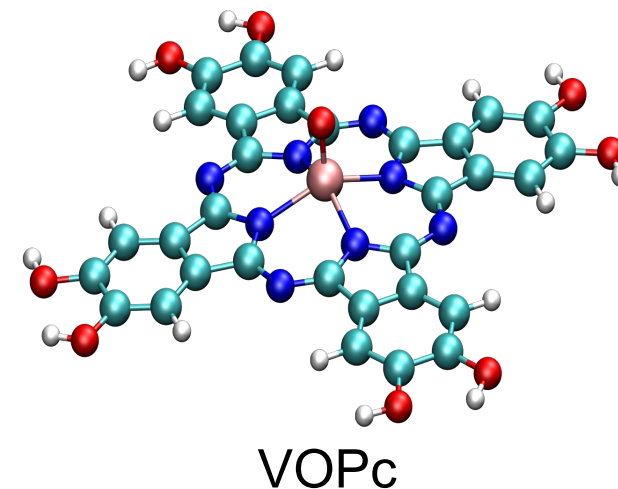


Unpublished.

Parameters taken from Roman Dmitriev *et al.* *J. Chem. Phys.* **2025**, 163, 224110

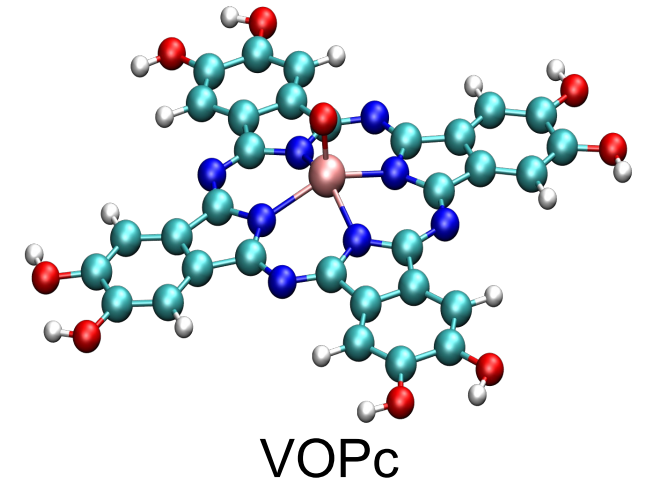
Bottleneck

$$H_{\text{sph}} = \mu_B \mathbf{B} \cdot \left[\sum_{\alpha \mathbf{q}} \left(\frac{\partial \mathbf{g}}{\partial Q_{\alpha \mathbf{q}}} \right) Q_{\alpha \mathbf{q}} + \sum_{\substack{\alpha, \beta \\ \mathbf{q}, \mathbf{q}'}} \left(\frac{\partial^2 \mathbf{g}}{\partial Q_{\alpha \mathbf{q}} \partial Q_{\beta \mathbf{q}'}} \right) Q_{\alpha \mathbf{q}} Q_{\beta \mathbf{q}'} + \dots \right] \cdot \boldsymbol{\sigma}$$



Bottleneck

$$H_{\text{sph}} = \mu_{\text{B}} \mathbf{B} \cdot \left[\sum_{\alpha \mathbf{q}} \left(\frac{\partial \mathbf{g}}{\partial Q_{\alpha \mathbf{q}}} \right) Q_{\alpha \mathbf{q}} + \sum_{\substack{\alpha, \beta \\ \mathbf{q}, \mathbf{q}'}} \left(\frac{\partial^2 \mathbf{g}}{\partial Q_{\alpha \mathbf{q}} \partial Q_{\beta \mathbf{q}'}} \right) Q_{\alpha \mathbf{q}} Q_{\beta \mathbf{q}'} + \dots \right] \cdot \boldsymbol{\sigma}$$



- Challenges

- Electronic structure: treatment of spin-orbit coupling
- Coupling: Finite difference calculation $O(N^2)$, 13224 single points for VOPc!

- potential AI acceleration?

- Weak couplings: $\mathbf{g}$ -tensor is a “flat” function of geometry
- What architecture? Transformer, MACE, ...
- How to evaluate gradient? Finite difference, automatic differentiation?

Summary

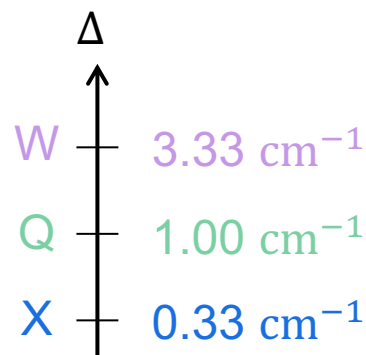
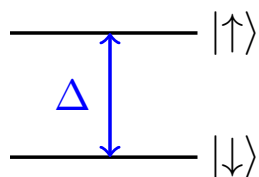
- A minimal open quantum system model for spin-1/2 relaxation
 - Analytical FGR rate expressions
 - Numerically exact eDEOM
- Applications
 - Materials with strong spin-phonon coupling
 - Materials with complex vibrational environment
- Challenges
 - The accuracy of electronic structure theory
 - Steep computation cost

Thank you for your attention

The model Hamiltonian and the choice of parameters

- Two-level system

$$H_s = \frac{\Delta}{2} \sigma_z$$



Thermal energy of 5 Kelvin :
3.5 cm⁻¹

- Spectral density for spin-phonon coupling

$$H_{\text{ph}} = \sum_k \omega_k b_k^\dagger b_k$$

$$Q \equiv \sum_k c_k \frac{b_k^\dagger + b_k}{\sqrt{2}}$$

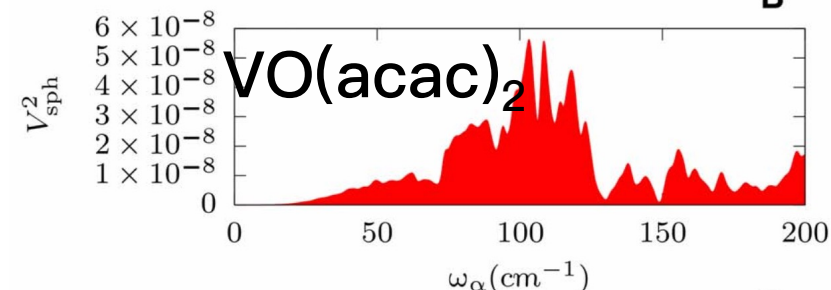
$$J(\omega) = \frac{\pi}{2} \sum_k c_k^2 \delta(\omega - \omega_k)$$



Collective mode Q

Spectral density
For spin-phonon couplings

Realistic spectral densities



Lunghi, A.; and Sanvito, S. *Sci. Adv.* **2019** 5 (9)

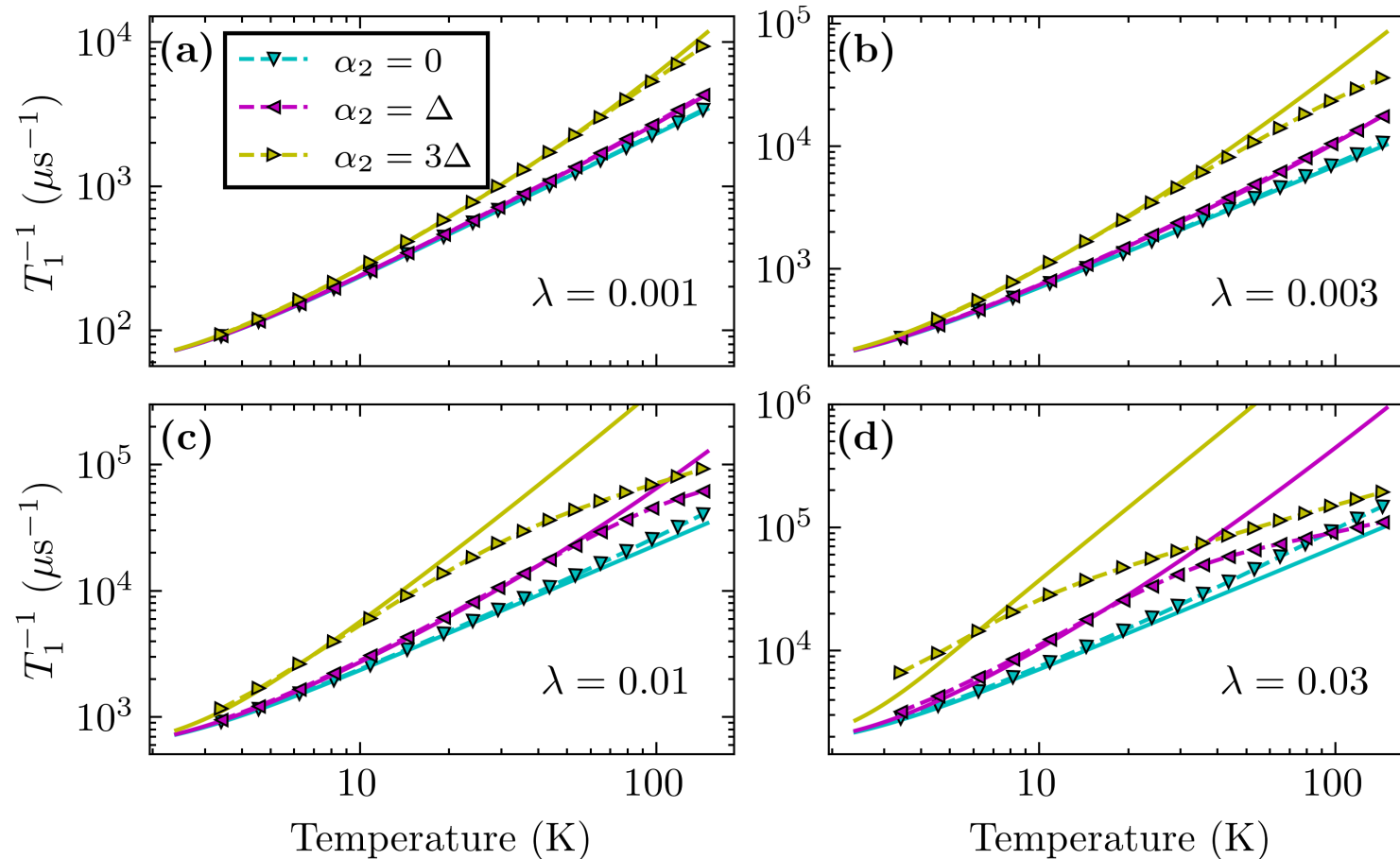
Analytical toy spectral densities:

- Ohmic, Drude, ...

The failure of power laws in strong coupling

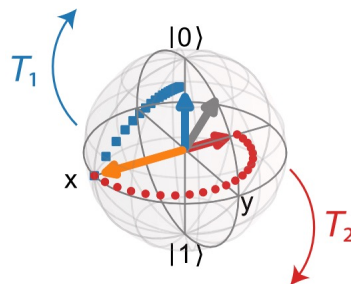
$$H_{\text{sph}} = (\alpha_1 \hat{Q} + \alpha_2 \hat{Q}^2) \sigma_x$$

FGR: solid line
DEOM: symbolled lines

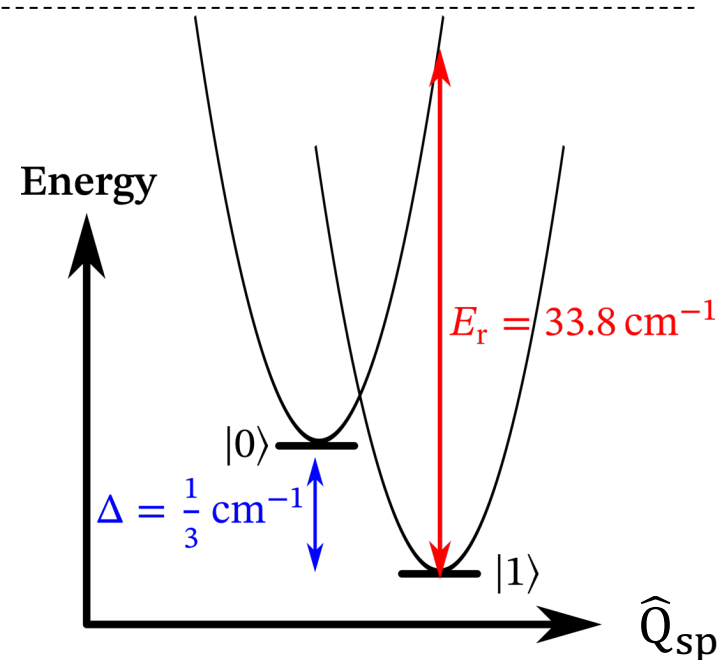
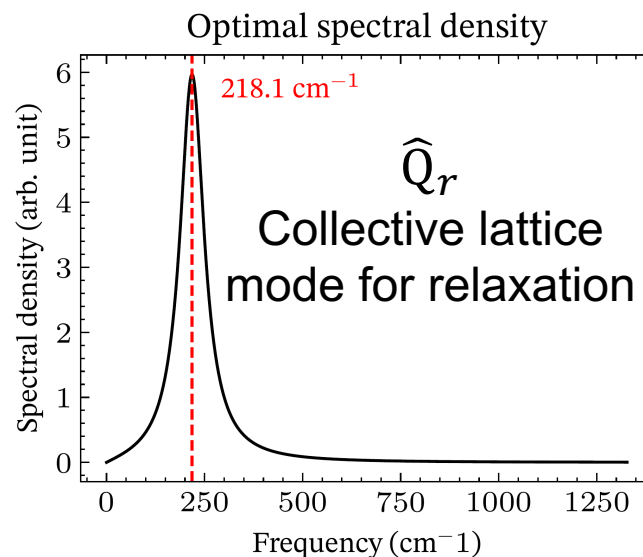
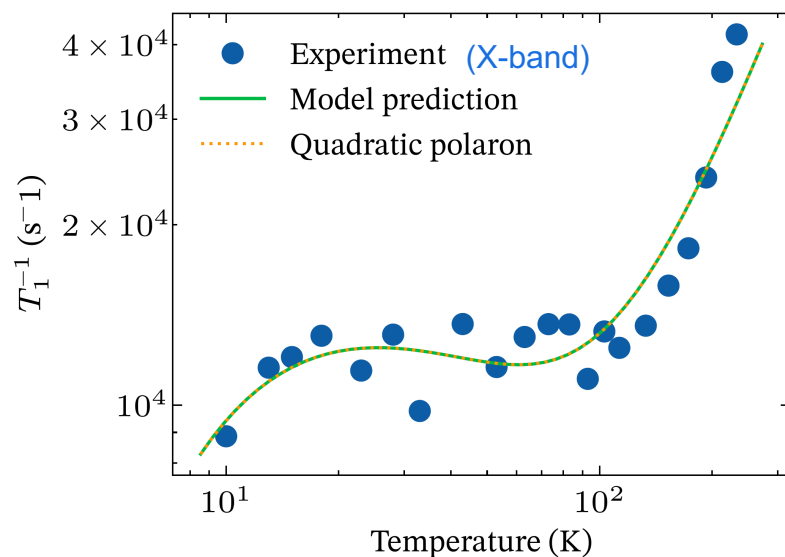


Fitting the T_1 temperature of $\text{Zn}_3(\text{HOTP})$ via “spin-phonon polaron”

Off-diagonal
coupling channel



dominant
“Spin-phonon polaron”



$$J(\omega) = \frac{2\lambda\zeta\omega_B^2\omega}{(\omega^2 - \omega_B^2)^2 + \zeta^2\omega^2}$$

ω_B (cm ⁻¹)	ζ (cm ⁻¹)	ω_{max} (cm ⁻¹)	E_r (cm ⁻¹)
221.2	74.7	218.1	33.8